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Research article

LSTM stock prediction model based on blockchain

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ABSTRACT

The stock market is a vital component of the financial sector. Due to the inherent uncertainty and volatility of the stock market, stock price prediction has always been both intriguing and challenging. To improve the accuracy of stock predictions, we construct a model that integrates investor sentiment with Long Short-Term Memory (LSTM) networks. By extracting sentiment data from the "Financial Post" and quantifying it with the Vader sentiment lexicon, we add a sentiment index to improve stock price forecasting. We combine sentiment factors with traditional trading indicators, making predictions more accurate. Furthermore, we deploy our system on the blockchain to enhance data security, reduce the risk of malicious attacks, and improve system robustness. This integration of sentiment analysis and blockchain offers a novel approach to stock market predictions, providing secure and reliable decision support for investors and financial institutions. We deploy our system and demonstrate that our system is both efficient and practical. For 312 bytes of stock data, we achieve a latency of 434.42 ms with one node and 565.69 ms with five nodes. For 1700 bytes of sentiment data, we achieve a latency of 1405.25 ms with one node and 1750.25 ms with five nodes.

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1. Introduction

In today's rapidly evolving financial markets, stock price prediction has always been an important issue of concern for investors and researchers [1]. Stock market prediction is crucial for investor decision-making. Accurate price predictions can help investors make informed investment choices in a complex market environment, thereby reducing risks and maximizing returns [2]. With advancements in AI technology, particularly the rise of deep learning, methods for analyzing and predicting the stock market are continuously evolving [3]. Traditional forecasting methods often rely on linear models and fundamental analysis. However, with the dramatic increase in data volume and heightened market volatility, these methods face numerous challenges [4]. Deep learning, especially Long Short-Term Memory (LSTM) networks, has gradually become a popular choice for financial market predictions due to their excellent time series modeling capabilities [5]. In addition, the development of blockchain offers new solutions for data security and transparency [6–9]. Therefore, this paper aims to explore an LSTM-based stock price prediction model and deploy it on the blockchain to enhance its reliability and traceability.

LSTM in Stock Prediction. As a branch of machine learning, deep learning utilizes multi-layer neural networks to extract features from vast amounts of data and perform pattern recognition [10]. In recent years, LSTM, a special type of RNN, has gained significant attention due to its superior performance in handling time series data. By introducing a gating mechanism, LSTM effectively addresses the issues of gradient vanishing and explosion that traditional RNNs face during long sequence training [11]. By capturing long-term dependencies in time series data, LSTM can learn complex market patterns, providing more accurate results for stock price prediction [12]. In stock prediction, the input for LSTM typically includes historical price data, trading volume, technical indicators, and other information [13]. However, relying solely on historical stock data for predictions is insufficient, as market sentiment is also a significant factor influencing stock prices. Investor sentiment reflects the market's expectations for future trends, and sentiment analysis can provide additional contextual information to the model, thereby enhancing prediction accuracy [14]. By combining historical price data with sentiment analysis features, we use the LSTM model to achieve a higher accuracy in the predictions.

Blockchain. In the context of increasing emphasis on data security and transparency, the introduction of blockchain offers new solutions for the financial sector [15]. Blockchain is a decentralized distributed ledger technology characterized by immutability, transparency, and traceability [16]. These features

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make the blockchain an ideal platform for storing and sharing financial data. Deploying LSTM-based stock prediction models on the blockchain can effectively enhance the security and credibility of data. Firstly, the immutability of the blockchain ensures the integrity of the input data, preventing malicious tampering [17,18]. Secondly, the transparency of the blockchain allows all participants to access and verify data in real time, thereby enhancing the credibility of the model [19,20]. Furthermore, the decentralized nature of the blockchain reduces the risk of single points of failure and improves the stability of the system [21,22]. By combining the LSTM model with blockchain, we aim to develop a more secure and more transparent stock price prediction system. This system will not only provide relatively accurate prediction results but also ensure the reliability of the data and the security of the system, thereby better supporting investors in making informed decisions.

Our contributions. We make the following contributions:

- We incorporated investor sentiment factors as a feature in our LSTM-based stock prediction model and innovatively used news from the Financial Post website as our dataset. By analyzing sentiment data along with stock trading indicators, we significantly enhanced the model's sensitivity to market fluctuations and predictive accuracy. We achieved an MAE of 0.01996 and an RMSE of 0.02542, compared to the traditional LSTM model, which recorded an MAE of 0.0319 and an RMSE of 0.0409. This underscores the importance of sentiment factors in financial decision-making.
- We employed blockchain to ensure data security. The decentralized nature and immutability of blockchain effectively prevent data tampering or loss, ensuring data integrity during transmission and storage. This provides a higher level of security for financial management.
- By deploying the stock prediction system on the blockchain, we improved the overall security of the system. The distributed characteristics of blockchain reduce the risk of single points of failure, enabling the system to maintain greater stability and robustness in the face of malicious attacks. We also ensure that the system is transparent and tamper-proof. The blockchain allows all operations and data records to be publicly accessible, which enhances our system's credibility. This has significant practical implications for the efficiency and security of financial systems.
- Through sentiment analysis, LSTM networks, and blockchain, we have achieved an accurate and efficient online stock prediction model. For stock data, we achieve a latency of 434.42 ms with one node and 565.69 ms with five nodes. For sentiment data, we achieve a latency of 1405.25 ms with one node and 1750.25 ms with five nodes. The successful implementation of this model provides more precise and secure decision support for investors and financial institutions.

The structure of the rest of the paper is as follows: we first discuss the related work in Section 2. In Section 3, we present our methodology, followed by the experimental results in Section 4. Finally, we conclude our work in Section 5.

2. Related work

2.1. Stock prediction work

With the development of deep learning, research on stock prediction has shifted from traditional linear mathematical models to more accurate deep learning-based approaches. Traditional Recurrent Neural Network (RNN) models used for handling sequential data face issues of vanishing and exploding gradients, making it challenging to train long sequences [23]. As a result, researchers do not frequently rely solely on RNN models for stock prediction. To address these issues, several improved variants of RNN have emerged, such as Long Short-Term Memory (LSTM) networks and Gated Recurrent Units (GRUs), which enhance memory retention and gradient propagation through the introduction of gating mechanisms [24]. Among these, LSTM has been widely used for stock price prediction. Siami-Namini et al. [25] compared LSTM and ARIMA for time series forecasting, and the results showed that the root mean square error (RMSE) of LSTM predictions was significantly lower than that of ARIMA predictions. Bathla and Gourav [26] compared LSTM with SVR using various stock indexes data, such as S&P 500, NYSE, NSE, BSE, NASDAQ, and Dow Jones Industrial Average. Experimental analysis demonstrated that LSTM provides better accuracy compared to SVR.

To improve prediction accuracy, some researchers have combined LSTM with other models. Buachuen et al. [27] proposed a novel approach that integrates advanced machine learning techniques to predict stock prices. The study explored deep learning models, including LSTM–CNN and CNN–LSTM, as well as attention layers, to capture and emphasize important temporal features in stock data. The LSTM–CNN hybrid model demonstrated excellent performance in capturing long-term temporal dependencies in stock patterns. Wu et al. [28] proposed a new framework structure called SACLSTM, which is based on graphs and integrates CNN and LSTM, thereby improving prediction performance compared to previous methods. Tian et al. [29] introduced a hybrid model named LSTM-BO-LightGBM, which combines an LSTM network based on Pearson's correlation coefficient with Bayesian-optimization LightGBM to address the issue of stock price volatility prediction.

Some researchers have also made enhancements in feature engineering. Chen et al. [30] utilized genetic algorithms (GA) for feature selection and then employed the LSTM model for stock prediction, improving the prediction accuracy by selecting appropriate feature parameters. Jin et al. [31] proposed a stock market prediction model based on deep learning, which takes into account investors' sentiment tendencies and employs Empirical Mode Decomposition (EMD) to progressively decompose the complex sequences of stock prices, resulting in improved prediction accuracy. To better address stock prediction issues, Ji et al. [32] proposed the DOC-W-LSTM model, which is based on sentiment analysis and the LSTM model, and it predicts the stock prices of Chinese healthcare companies.

2.2. Blockchain application work

Blockchain utilizes various computer technologies, including distributed data storage, consensus mechanisms, and cryptographic algorithms [33]. Blockchain is a decentralized, tamper-proof, and efficient distributed ledger that is a disruptive technology to traditional systems, significantly enhancing security, and is widely applied in the financial sector [34].

Currently, blockchain has been applied in multiple financial fields, including insurance and banking [35]. Implementing blockchain in the insurance industry brings numerous benefits, such as improved efficiency, reduced costs, enhanced security, and lower risks [36]. The applications of blockchain in the insurance sector mainly include smart contracts and claims processing, data security and privacy protection, and fraud detection [37]. Bhadra et al. [38] introduced a novel solution for the subrogation process in the automotive insurance industry based on blockchain, streamlining the auto insurance process, eliminating untrustworthy intermediaries, and enhancing security. Zhou et al. [39] proposed a blockchain-based medical insurance storage system called MIStore, which provides high credibility for users and is deployed on the Ethereum blockchain, demonstrating good

performance. Lepoint et al. [40] presented BlockCIS, a blockchain-based system for continuous monitoring and processing in cyber insurance, achieving an automated, real-time, and immutable feedback loop among insurance companies, clients, third parties, and potential auditors. Loukil et al. [41] proposed CioSy, a blockchain-based collaborative insurance system designed to monitor and process insurance transactions, aiming to automate policy processing, claims handling, and payments through smart contracts, with an experimental prototype built on the Ethereum blockchain. The application of blockchain in the insurance industry can improve operational methods, enhance efficiency, reduce costs, and provide better security and privacy protection, thus attracting an increasing number of insurance companies and researchers to actively explore its applications [42].

The decentralization of blockchain allows it to establish credit mechanisms in situations where there is a lack of mutual trust among parties, effectively addressing some existing issues in the banking sector, reducing costs, and enhancing security [43]. Blockchain is primarily applied in banking for cross-border payments and remittances, asset management, and loans and credit [44]. Daluwathumullagamage and Sims [45] conducted a systematic literature review of work from 2013 to 2020 to explore the applications of blockchain in the banking industry and analyze key themes, trends, discrepancies, and gaps from both academia and industry practitioners. Additionally, a blockchain framework and lifecycle applicable to banking were developed. Mamun et al. [46] proposed a banking management system architecture that integrates artificial intelligence, blockchain, and Hyperledger as part of a multi-layer security and management system, enhancing the security of banks. Xu et al. [47] introduced a blockchain-based data sharing model (PPM) for open banking, utilizing programmable smart contracts as intermediaries between users and third-party services, modifying the data layer, smart contract layer, and application layer. Experiments demonstrated that this model is a secure and reliable system for open banking.

Blockchain offers the banking sector more efficient, secure, and transparent solutions, improving various business processes and enhancing customer experiences [48]. Both the banking industry and researchers have been studying the application of blockchain in banking, achieving certain progress.

Currently, blockchain has been applied in multiple financial fields, including banking and insurance, but it has not yet been utilized in stock prediction [49]. We deploy the stock prediction model on the blockchain, effectively enhancing the security of the prediction system and reducing accuracy loss due to malicious attacks.

3. Our system

An overview of our system is shown in Fig. 1. In the stock prediction system, our stock data and news data come from Quandl and Financial Post respectively. Stock data refers to the historical price and trading volume of stocks, and news data is the financial news information crawled from the Financial Post. In the data processing layer, we perform basic preprocessing on the stock data and use the Vader sentiment dictionary to quantify the news data to obtain the quantified sentiment data. Then, the stock data and the quantified sentiment data are merged by date to get a full dataset. Next, in the prediction layer, we input the full dataset into the LSTM model for training, and combine the stock data and sentiment index for prediction. Finally, we deploy the stock prediction system on the blockchain. The blockchain layer is used for data storage to ensure the security and immutability of data and systems.

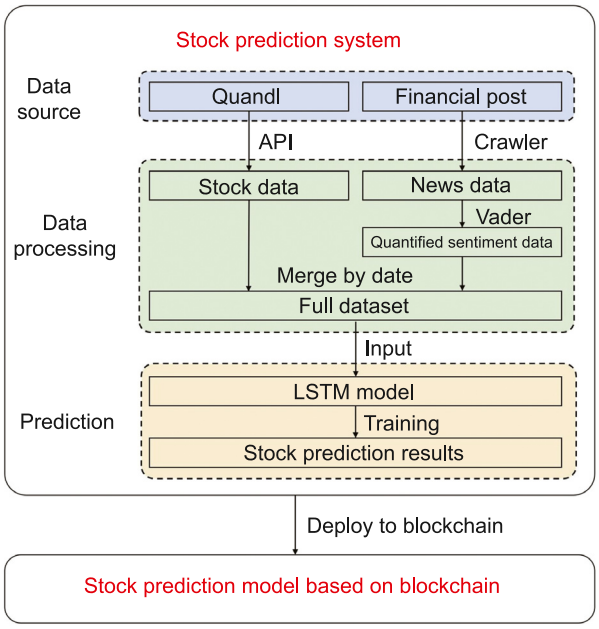


Fig. 1. System overview.

3.1. Dataset

To achieve effective stock price forecasting, we obtain high-quality historical stock price data and relevant investor sentiment data from reliable sources.

We chose to use Quandl as the primary source for stock price data. Quandl is a widely used financial data platform that offers various types of market data, including stocks, bonds, futures, and economic indicators [50]. It has a broad coverage of data, a high update frequency, and provides a convenient API for users to retrieve data. We selected a number of representative stocks on Quandl, including Microsoft Corporation (MSFT), American Express (AXP), Apple Inc. (AAPL), Boeing Company (BA), Caterpillar Inc. (CAT), Cisco Systems (CSCO), and Chevron Corporation (CVX), among others. These stocks include large technology companies, energy companies, and financial institutions, covering different industries and markets to ensure the model's generalization capability. By writing Python scripts and utilizing Quandl's API, we could specify the desired data fields, namely the date, stock ticker, and closing price. We chose the stock data from January 1, 2010, to December 31, 2019, to capture both long-term trends and short-term fluctuations in the market.

To analyze the impact of investor sentiment on stock prices, we chose the Financial Post website as the primary source of sentiment data. Financial Post is a media outlet that provides financial news and market analysis, covering the latest market trends, company news, and economic developments.

We used web scraping techniques to extract relevant financial news articles from the Financial Post website. The selected articles include market analyses, company earnings reports, and economic data releases, which typically trigger fluctuations in investor sentiment. The data we collected consists of the publication date, title, and content of the articles for subsequent sentiment analysis. We organized the results into a DataFrame containing news dates, text, and links, filtering out entries that lacked text.

We used the Vader (Valence Aware Dictionary and sEntiment Reasoner) sentiment analysis tool to extract investor sentiment. Vader is designed explicitly to analyze sentiment in social media texts and can effectively identify emotional tendencies within the

text. For each scraped news article, we first segmented the text and then used Vader for sentiment analysis, obtaining scores for positive, negative, and neutral sentiments. Positive text tends to have a composite score close to 1; negative text has a composite score close to −1; and neutral text has a composite score close to 0. The generated sentiment data will be stored in a time series format, facilitating alignment with stock price data. Finally, we will integrate the stock price data obtained from Quandl with the sentiment data scraped from the Financial Post. We will align the sentiment scores with the corresponding stock price data using timestamps, ensuring that each time point has corresponding sentiment data.

3.2. Sentiment analysis

We use Vader to quantify investor sentiment. Vader is a rule-based sentiment analysis tool proposed by Hutto and Gilbert at the AAAI conference in 2014 [51]. It performs sentiment recognition in text by constructing a sentiment lexicon, making it particularly suitable for social media text analysis. Vader can handle informal language, abbreviations, emojis, and other internet slang. It matches words in the text with the sentiment lexicon, calculating a sentiment score for each word. The overall sentiment score of a sentence is then computed based on the scores of the matched sentiment words, the context of the words (such as the presence of negation), and other rules. A score greater than 0 indicates positive sentiment, a score less than 0 indicates negative sentiment, and a score close to 0 indicates neutral sentiment.

Vader has long been regarded as the gold standard for dictionary-based sentiment analysis, applicable to various types of text content, and it possesses several unique advantages [52]. Vader has advantages in terms of simplicity and ease of use, the ability to handle informal text, context understanding, computational efficiency, and interpretability of results. Therefore, in this social media context, we choose Vader for sentiment analysis.

3.3. LSTM model

We employ Long Short-Term Memory (LSTM) networks to construct a stock price prediction model. As a special type of Recurrent Neural Network (RNN), LSTM excels in handling time series data and can capture long-term dependencies within the data. In this section, we will discuss in detail the model architecture, hyperparameter settings, and the selection of loss functions and optimization algorithms to ensure the model's effectiveness and accuracy.

3.3.1. LSTM model architecture

The input layer of the model receives the processed time series data. Our input data includes historical stock prices and quantified investor sentiment data, which have been appropriately preprocessed and normalized. The data must comply with the requirements of LSTM, specifically in a three-dimensional array format (number of samples, time steps, number of features). We set the time step to 7, meaning that the model will make predictions based on the data from the past 7 days to capture the trends in price changes.

The core of the model is the LSTM layer. The LSTM layer contains 200 neurons, a choice made based on the complexity of the data. The number of LSTM neurons directly affects the model's learning capability; too few neurons may fail to capture important patterns in the data, while too many could lead to overfitting. 200 neurons is a suitable choice, allowing the model to learn the important features within the time series effectively.

To prevent the model from overfitting, we added a Dropout layer after the LSTM layer. Dropout is a regularization technique that reduces the model's dependence on the training data by randomly dropping a certain proportion of neurons. We set the Dropout rate to 20%. This setting aims to enhance the model's generalization ability, ensuring more stable performance on unseen data.

The final layer of the model is the Dense layer, which outputs a single value representing the predicted stock price. The role of the Dense layer is to map the features extracted by the LSTM layer to the final prediction result. In this way, the model can convert the learned time series features into specific price predictions.

3.3.2. Hyperparameter settings

The setting of hyperparameters is crucial for the performance of the model. We determine the hyperparameters of the model, including learning rate, batch size, number of LSTM units, and time steps. We use grid search to optimize hyperparameters.

We set the number of training epochs to 300. The number of training epochs indicates how often the model learns from the training data. An appropriate number of epochs ensures that the model adequately learns the features in the data, while too many epochs may lead to overfitting.

The batch size is set to 25, meaning that the model will use 25 samples for training during each weight update. The batch size affects the model's stability and convergence performance. A smaller batch size typically results in longer training times but provides more frequent updates, allowing the model to better adapt to changes in the data. Conversely, a larger batch size can utilize the parallel computing capabilities of GPUs more effectively, thus improving training speed, but it may also increase the risk of overfitting. We chose 25 as a balanced point to ensure the model's performance.

The dropout rate is another important hyperparameter. We chose a dropout rate of 20%, which means that 20% of the neurons will be randomly ignored during each training iteration. The dropout technique introduces randomness during the training process, reducing the model's reliance on specific neurons and effectively preventing overfitting.

The learning rate is an important parameter that controls the speed of weight updates in the model. If the learning rate is too high, it may cause the model to oscillate during training or even fail to converge. Conversely, if the learning rate is too low, it may result in excessively slow training or cause the model to get stuck in a local minimum. We used Adam's default learning rate, which yielded good results.

Finally, to prevent overfitting, we implemented an early stopping mechanism during the model training process. We monitor the changes in validation loss. If the validation loss does not improve for 10 consecutive training epochs, the training will automatically stop. This mechanism effectively reduces unnecessary training time and prevents the model from overfitting.

3.3.3. Loss function and optimization algorithm

The loss function measures the gap between the model's predicted values and the true values. We use Mean Squared Error (MSE) as the loss function. MSE effectively reflects the model's predictive performance by calculating the squared differences between predicted and true values. One of the advantages of MSE is its pronounced penalty for more significant errors, which encourages the model to pay more attention to samples with significant prediction deviations during training. As the training progresses, the value of the loss function gradually decreases, indicating that the model is continuously learning and optimizing. We assess the model's learning effectiveness and generalization ability by monitoring the changes in MSE on the validation set.

We utilize the Adam optimizer for training due to its adaptive learning rate characteristics, which allow it to dynamically

adjust the learning rate based on the gradient information of each parameter. The Adam optimizer performs exceptionally well when handling complex time series data, accelerating the model's convergence speed and improving prediction accuracy.

3.4. Deploy to blockchain

We deployed the LSTM stock prediction model, which incorporates investor sentiment, onto the blockchain to ensure data security, transparency, and system robustness.

3.4.1. Blockchain platform

We chose the Tian-Shu blockchain as a representative implementation, which is an underlying system independently developed by Yangtze Delta Region Institute of Tsinghua University. One could use other blockchains as well. Tian-Shu Blockchain is a distributed ledger based on blockchain that employs the Practical Byzantine Fault Tolerance (PBFT) algorithm as its consensus mechanism. PBFT is a fault-tolerant consensus algorithm primarily used to ensure that the system maintains consistency and correctness even in the presence of malicious nodes [53].

Tian-Shu Blockchain leverages the PBFT consensus mechanism to reach consensus in a relatively short time, making it suitable for applications requiring quick transaction confirmations, such as financial transactions. By tolerating malicious nodes, Tian-Shu Blockchain can maintain data consistency and integrity even when untrustworthy nodes exist within the network. Furthermore, Tian-Shu Blockchain supports various nodes joining and exiting, offering good scalability to adapt to evolving business needs. It also employs permission management and data encryption techniques to protect user privacy and sensitive data.

Therefore, considering the efficiency, security, and scalability of Tian-Shu Blockchain, we deployed the sentiment embedding LSTM stock prediction model on Tian-Shu Blockchain to achieve a more secure and efficient stock prediction system.

3.4.2. Data storage

We store stock data and sentiment data on the Tian-Shu Blockchain, leveraging the immutability and transparency of blockchain to ensure data security.

First, we upload the stock data and investor sentiment data to the Tian-Shu Blockchain. Each piece of data generates a unique hash value, which is recorded on the blockchain to ensure data integrity and immutability. During the data onboarding process, we utilize the blockchain's consensus mechanism to verify the legality and authenticity of each data entry. This process involves the validation and confirmation of nodes, preventing the insertion of malicious data. Users can query the data stored on the blockchain, ensuring that all participants have access to a transparent and trustworthy data source.

3.4.3. Security assurance

The decentralized nature of blockchain reduces the risk of single points of failure, as data storage and processing do not rely on a single node. We implement data redundancy and distributed validation. Data redundancy refers to the replication and storage of data across multiple nodes, ensuring that the system can continue to operate normally even if some nodes fail. Distributed validation involves multiple nodes within the blockchain network collectively verifying the legitimacy of the data, preventing any single node from tampering with the data.

To protect the data and the system, we have established strict access control mechanisms. Different access permissions are set based on the roles of various users, ensuring that only authorized individuals can access specific data and functionalities. We utilize Public Key Infrastructure (PKI) [54] to authenticate user identities, ensuring the security and traceability of the system.

We employ encryption technology to ensure the security of data stored on the blockchain. During data transmission, we use SSL/TLS protocols to encrypt the data, protecting against man-in-the-middle attacks and data breaches [55]. Additionally, we implement a secure key management system to ensure the safe storage and access of encryption keys, preventing key leakage.

4. Implementation and evaluation

In this chapter, we use Microsoft's stock prediction results as an example to conduct a detailed comparative analysis of the prediction outcomes of the LSTM model that incorporates investor sentiment versus the standard LSTM model. We divided the entire dataset into a training dataset and a testing dataset, with proportions of 70% and 30%, respectively. The training dataset is used for model training, while the testing dataset is used to evaluate the model's predictive ability on stock price. Finally, through an in-depth discussion of the experimental results, we analyze the impact of investor sentiment on stock price predictions.

4.1. Performance indicator

In deep learning models, selecting appropriate evaluation metrics is crucial for model assessment and optimization. Evaluation metrics reflect the performance of the model and guide the direction of model improvement. To comprehensively evaluate the model's predictive performance, we selected multiple evaluation metrics to analyze the model's performance from different perspectives. The following are the primary evaluation metrics we used:

Root Mean Square Error (RMSE): RMSE is a commonly used metric for measuring the difference between predicted values and true observed values. It reflects the model's predictive accuracy by calculating the average of the squared prediction errors and then taking the square root. The smaller the RMSE, the more accurate the model's predictions are. The formula for calculating RMSE is the following:

$$\mathrm{RMSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2},$$

where y is the true value, $\hat{y}$ is the predicted value and n is the number of samples. The advantage of RMSE is that it gives higher weight to larger errors, which means it can effectively reflect deficiencies in model performance in certain cases.

Mean Absolute Error (MAE): Mean Absolute Error (MAE) is another commonly used performance indicator that calculates the average absolute differences between predicted values and true values. The formula for calculating MAE is:

$$\mathrm{MAE} = \frac{1}{n}\sum_{i=1}^{n}|y_i - \hat{y}_i|.$$

The advantage of MAE lies in its intuitiveness and ease of understanding, as it directly represents the magnitude of the prediction errors. Unlike RMSE, MAE applies a linear penalty to errors, which makes it often more robust when dealing with datasets that contain outliers.

We used these metrics to compare the predictive performance of different models and analyze the impact of investor sentiment on the model outcomes.

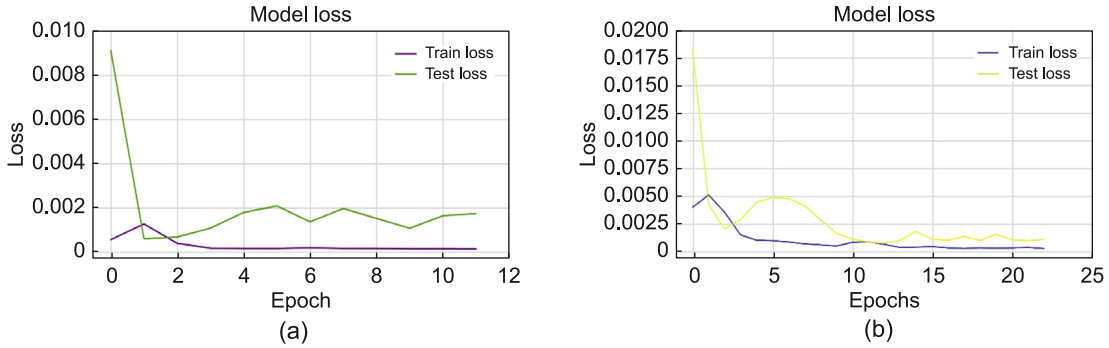


Fig. 2. Loss values of two LSTM models. (a) Loss of LSTM model. (b) Loss of Sentiment embedding LSTM model.

Table 1
Value of performance indicators.

Model	Dataset	MAE	RMSE
LSTM	Training	0.0078	0.01028
LSTM	Test	0.0319	0.0409
Sentiment Embedding LSTM	Training	0.01179	0.01423
Sentiment Embedding LSTM	Test	0.01996	0.02542

Note: the results of the RMSE and MAE metrics for both models in the training dataset and the test dataset.

4.2. Model performance indicator value

We constructed two LSTM-based models for stock price prediction: one is an LSTM model that does not include investor sentiment features, and the other incorporates investor sentiment features.

We calculated the RMSE and MAE metrics for both models in the training data set and the test data set, and the results in the test data set are shown in Table 1. MAE measures the average absolute difference between predicted and actual values, and a smaller value indicates higher predictive accuracy of the model. RMSE is the average of the square roots of the prediction errors and is more sensitive to large errors. From the data in the Table 1, it can be seen that the sentiment embedding LSTM model outperforms the LSTM model without sentiment features in both MAE (0.01996) and RMSE (0.02542). In comparison, the LSTM model without sentiment features has an MAE of 0.0319 and an RMSE of 0.0409. This indicates that the introduction of investor sentiment features significantly enhances the predictive performance of the model.

4.2.1. LSTM model

In the LSTM model without sentiment features, the performance indicator for the training dataset showed an MAE of 0.0078 and an RMSE of 0.01028. The MAE for the test dataset is 0.0319, and the RMSE is 0.0409. These values indicate that the model performs well on the training dataset but exhibits relatively more significant errors on the test dataset, particularly with the RMSE value approaching 0.041, which highlights significant prediction errors.

Regarding the loss values, Fig. 2(a) shows the variation of loss values for the LSTM model without sentiment features during training and testing. It is evident that although the model performs well on the training dataset, the loss values are considerably higher on the test dataset. The loss values on the test dataset showed significant fluctuations in the early training stages, but overall, there was a decreasing trend as the number of training epochs increased. However, the loss values on the test dataset remained relatively high throughout, ultimately failing to converge to a low, stable value, indicating inadequate predictive capability of the model.

4.2.2. Sentiment embedding LSTM model

The sentiment embedding LSTM model has MAE and RMSE values of 0.01179 and 0.01423, respectively, on the training set. The MAE for our model on the test dataset is 0.01996, and the RMSE is 0.02542. Compared to the LSTM model without sentiment features, the introduction of sentiment features has led to an increase in both MAE and RMSE on the training dataset. This phenomenon may indicate that the addition of sentiment features has increas()ed the complexity of the model, resulting in a decrease in its fitting capability on the training data. However, both MAE and RMSE have significantly decreased on the test dataset, especially the RMSE, which dropped from 0.041 to 0.025. This indicates a substantial improvement in the model's predictive capability.

The variation in loss values for the LSTM model with sentiment features is shown in Fig. 2(b). The loss values of this model decrease more rapidly in the early stages of training and ultimately stabilize at a level that is significantly lower than that of the LSTM model without sentiment features. This suggests that the introduction of sentiment features has significantly improved the model's learning efficiency and resulted in better predictive performance.

4.3. Prediction results

We used both the LSTM model without investor sentiment features and the LSTM model that incorporates investor sentiment features to predict MSFT stock data. The comparison between the predicted values and the true values is illustrated in Fig. 3.

4.3.1. LSTM model

Fig. 3(a) shows the prediction results of the LSTM model without sentiment features on the test dataset. By comparing the curves of the predicted values and the true values, we can see that the model can capture the overall upward trend, especially over longer time periods, where the predicted values and true values generally align in their direction of change. This indicates that the model can grasp trends in long time series.

However, during certain periods, such as mid-2017 and early 2018, there is a significant discrepancy between the predicted values and the true values, particularly in areas of sharp price fluctuations. This suggests that the model performs inadequately in dealing with short-term volatility and fails to capture instantaneous market changes effectively. Additionally, in some cases, the changes in the predicted values exhibit a certain lag compared to the true values. When the stock market experiences rapid increases or decreases, the model's response is slow, failing to adjust the predicted values on time. It is evident that using only stock price data as features in the LSTM model makes it challenging to predict stock prices accurately.

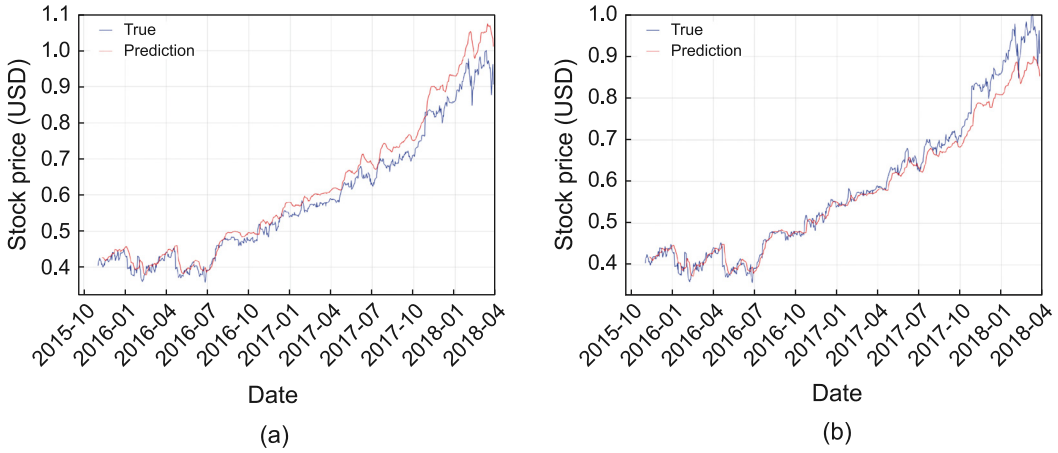


Fig. 3. Prediction results of two LSTM models. (a) Prediction results of LSTM model. (b) Prediction results of sentiment embedding LSTM model.

4.3.2. Sentiment embedding LSTM model

The prediction results of the sentiment embedding LSTM model on the test dataset are shown in Fig. 3(b). By comparing the curves of the predicted values and the true values, it is evident that the model's predictive performance has significantly improved, especially before 2019.

The gap between the predicted values and the true values of our model has narrowed, especially during periods of high price volatility, where our model is better able to capture instantaneous market changes. For instance, during the sudden increase in stock prices at the beginning of 2017, the sentiment embedding LSTM model was able to predict prices much more accurately compared to the LSTM model without sentiment features, resulting in a smaller discrepancy from the true values. The sentiment embedding LSTM model demonstrates enhanced responsiveness during rapid market changes. When unexpected events occur in the stock market, the model can adjust more quickly.

Experimental results indicate that incorporating sentiment features makes the model more sensitive to the dynamics of the stock market, which is closely related to the psychological expectations of investors provided by the sentiment features. Thus, the accuracy of predictions for stock data is improved.

In summary, the sentiment embedding LSTM model demonstrates lower loss values, reduced MAE and RMSE indicator values, and higher prediction accuracy. This underscores the importance of sentiment features in the financial market, indicating that the LSTM model with investor sentiment features performs better in stock price prediction.

4.4. Consensus latency and throughput

We deployed the prediction system on the blockchain and evaluated its on-chain performance. We used the m5.xlarge instance for our evaluation. The m5.xlarge instance is equipped with 4 virtual CPUs and 16 GB of memory. It provides up to 10 Gbps of network bandwidth, making it suitable for scenarios that require a balance between computational and memory demands. We evenly distributed the nodes across several different regions: us-west-2 (Oregon, US), us-east-2 (Ohio, US), ap-southeast-1 (Singapore), and eu-west-1 (Ireland). We set the client number to 10, and tested the consensus latency and throughput under different network sizes and batch sizes.

We conducted experiments under different network scales and batch sizes. Here, f represents the number of nodes, and b denotes the batch size, which refers to the number of transactions submitted to the blockchain in a single operation. For each experiment, we tested the consensus latency and the throughput. Consensus latency is the time required for nodes in the blockchain network to reach an agreement, while throughput refers to the number of transactions the system can process in a unit of time, typically measured in transactions per second (TPS). These metrics reflect the system's efficiency and processing capability.

We tested both stock data and sentiment data, which are 312 bytes and 1700 bytes in size, respectively. We evaluated the system's performance for cases where f was set to 1 and 5.

In the case of $f = 1$, we tested the stock data with a transaction size of 312 bytes and the sentiment data with a transaction size of 1700 bytes, with the results shown in Fig. 4(a). For the stock data, as the batch size increased, both consensus latency and throughput exhibited an overall upward trend. At a batch size of 100, the average consensus latency was 83 ms, and the throughput reached 663.57 TPS. When the batch size increased to 30,000, the latency rose to 434.42 ms, while the throughput increased to 34360.33 TPS, demonstrating the system's good performance under high load.

When testing the sentiment data with a larger transaction size, the latency increased compared to the stock data, but the system still performed well. The tests showed that at a batch size of 100, the latency was 102 ms, and the throughput was 525.76 TPS. As the batch size increased, the system managed the load effectively, ultimately resulting in a latency of 1405.25 ms and throughput of 12342.63 TPS at a batch size of 30,000. This indicates that the system possesses high efficiency and good processing capability for both stock data and sentiment data.

In the case of $f = 5$, we similarly tested the stock data with a transaction size of 312 bytes and the sentiment data with a transaction size of 1700 bytes, with the results shown in Fig. 4(b). It is evident that the system's consensus latency is generally higher than in the $f = 1$ scenario. This increase in latency can be attributed to the higher number of redundant replicas, which results in greater network communication overhead. However, the throughput still maintained a relatively high level to some extent, indicating that the system is capable of handling more concurrent requests. This suggests that while the added redundancy may lead to increased latency, it also enhances the system's ability to manage higher workloads efficiently.

In summary, under the PBFT consensus mechanism, the write performance for both the 312-byte stock data and the 1700-byte sentiment data demonstrated good results. Although the latency increased in the $f = 5$ scenario, the system still maintained a high throughput. These results indicate that the stock prediction

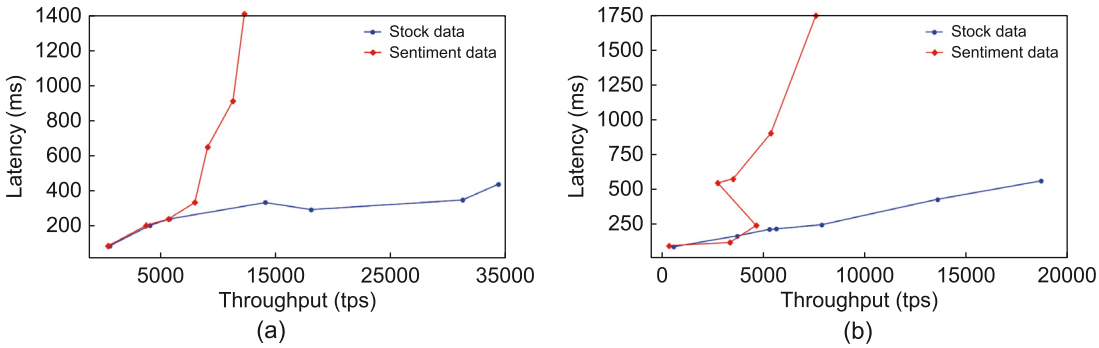


Fig. 4. Latency vs. Throughput. (a) The case of f = 1. (b) The case of f = 5.

system, when deployed on the blockchain, can achieve efficient data writing across different numbers of nodes and batch sizes, showcasing strong performance overall.

5. Conclusion

We combine blockchain with LSTM and incorporate investor sentiment analysis. We enhance the accuracy of stock predictions while fully leveraging the advantages of advanced machine learning techniques and blockchain to build a robust, secure, and transparent system. The results indicate that this hybrid approach not only improves prediction accuracy but also addresses key issues related to data security and system integrity.

By introducing investor sentiment analysis, we enhance the predictive capability of the LSTM model. We analyze investor sentiment through articles on financial websites and use the Vader sentiment lexicon to obtain quantifiable sentiment data. By adding this new feature of sentiment data, we successfully improve the accuracy of the LSTM model's predictions. Finally, we deploy the stock prediction system on the Tian-Shu Blockchain. By storing data on the blockchain, we ensure the immutability and verifiability of the prediction model inputs, significantly enhancing data credibility and improving system robustness. Additionally, the PBFT mechanism of the Tian-Shu Blockchain allows for rapid consensus among nodes, facilitating efficient processing. We conducted experiments on the system's latency, demonstrating that the prediction system exhibits low latency after deployment on the blockchain.

As a future direction, we could incorporate additional features beyond sentiment, such as financial statement indicators and industry indices, to further enhance the model's predictive capability. Furthermore, we can improve the LSTM neural network to strengthen its predictive performance.

CRediT authorship contribution statement

Yongdan Wang: Writing – review & editing, Writing – original draft, Methodology, Formal analysis. **Haibin Zhang:** Writing – original draft, Project administration, Funding acquisition, Conceptualization. **Baohan Huang:** Validation, Formal analysis, Data curation. **Zhijun Lin:** Resources, Writing – review & editing. **Chuan Pang:** Resources, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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